Student Exchange | DFD design atlas

Logical context, Level-0 and balanced lower-level views • website + adapter + API

Follow the data, one boundary at a time.

Prepared from the supplied PowerPoint design and the implemented Student Exchange website and backend. The model separates browser previews, server-owned identity data, persistent records and the independent listing API.

01 / Context and decomposition

A single context process expands into six major processes, then focused account, administration, access and maintenance views.

02 / Exact boundary accounting

Named data packages are preserved across every parent/child boundary. Composite flows split without adding or losing information.

03 / Editable and reproducible

Native editable PowerPoint objects, searchable vector PDF, separate SVGs, semantic JSON, Graphviz sources and automated structural checks.

18 discrete DFDs, 17 checked decomposition relationships and a full data dictionary.

Diagram index and reading path

The context uses process 0; Level-0 uses 1.0–6.0; child numbers identify their parent.

C0 Student Exchange
Context

L0 Student Exchange data flows
Level-0

P1 Marketplace preview data
Level-1 · parent 1.0

P2 Accounts and website sessions
Level-1 · parent 2.0

P3 Administration data
Level-1 · parent 3.0

P4 Access and forwarding data
Level-1 · parent 4.0

P5 Retained security data maintenance
Level-1 · parent 5.0

P6 Direct listing API data
Level-1 · parent 6.0

P21 Registration data transformation
Level-2 · parent 2.1

P22 Login token and cookie data
Level-2 · parent 2.2

P24 Saved profile and readback data
Level-2 · parent 2.4

P25 Logout and revocation data
Level-2 · parent 2.5

P32 Target user record management
Level-2 · parent 3.2

P34 Admin membership changes
Level-2 · parent 3.4

P35 Attempt inspection and clearing
Level-2 · parent 3.5

P42 Forwarding identity data
Level-2 · parent 4.2

P43 Application and IP observations
Level-2 · parent 4.3

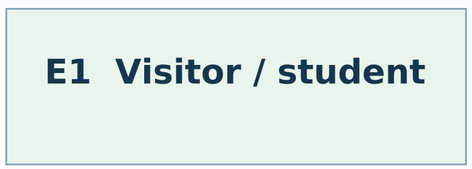
P44 User identity and attempt observations
Level-2 · parent 4.4

Use the focused child views to inspect details. Numbering identifies decomposition, not execution order.

Reference: DFD (1).pdf, pp. 6–9 and 15–23; README.md includes the complete index.

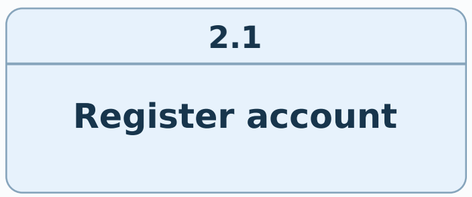
Four symbols, one consistent notation

Gane–Sarson style follows the reference: numbered rounded processes, entities, open-ended stores and arrows.



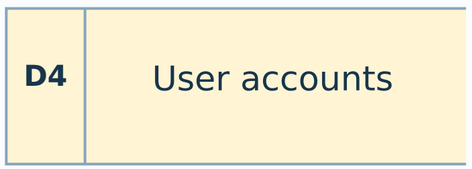
External entity

A person or outside system supplying or receiving data. Only E1, E2 and E3 are external to this whole-system model.



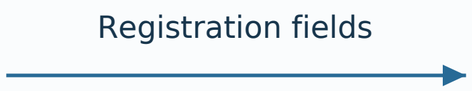
Numbered process

A verb phrase describes a transformation. Each internal process has data input and output. Parent numbers expand into child numbers.



Data store

A noun phrase names retained data. D1/D2 are browser preview records; D11/D12/D15 are process memory; SQLite stores are separate.



Directed data flow

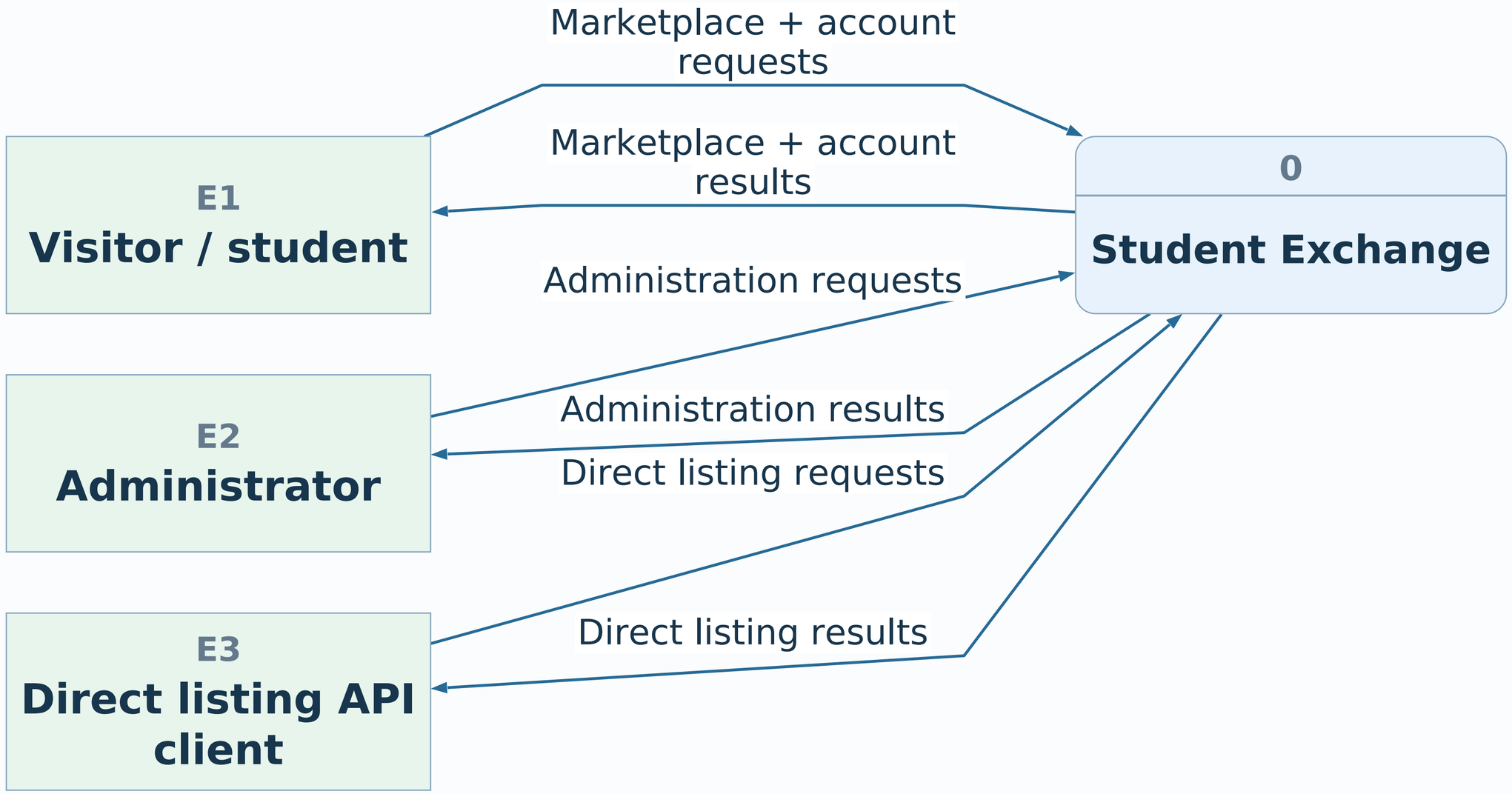
One arrow has one direction. Separate arrows represent reads and writes. Named composite packages are defined in the data dictionary.

Arrows carry noun-phrase data; they do not represent time, control, decisions, API call order or lifecycle transitions.

Reference: DFD (1).pdf, pp. 2–5, 10–14 and 24–29.

Context | Student Exchange

Context • Entire implemented system, with preview and listing limitations stated

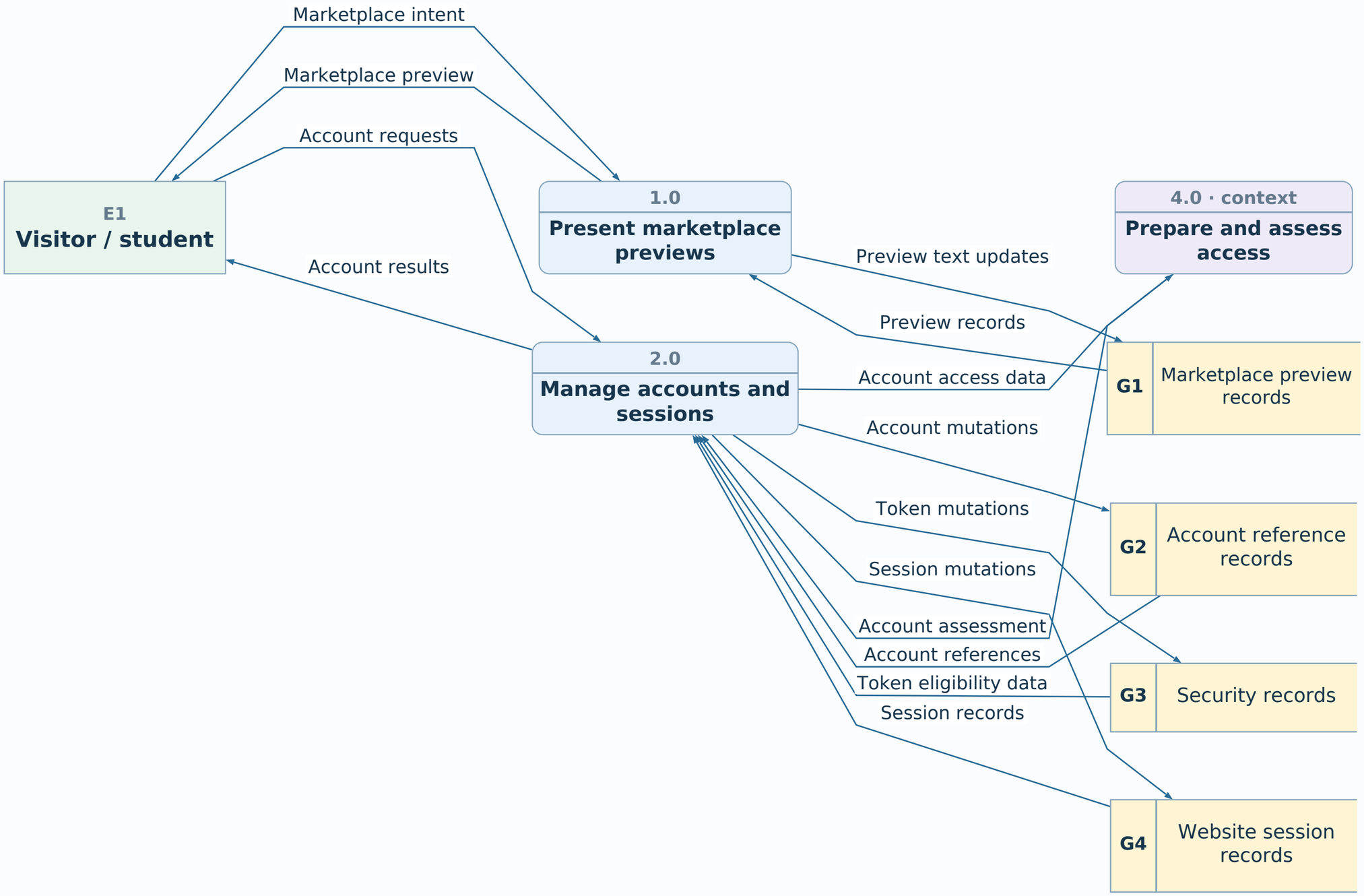


One system process, external entities and boundary data only. Internal stores do not appear.

Evidence: Website/src/App.tsx; server/api.mjs; Backend/Controllers/Listing.cs

Level-0 | Student Exchange data flows | 1/3

Level-0 • Website, Node adapter, backend and logical storage • same-level sheet for 1.0, 2.0

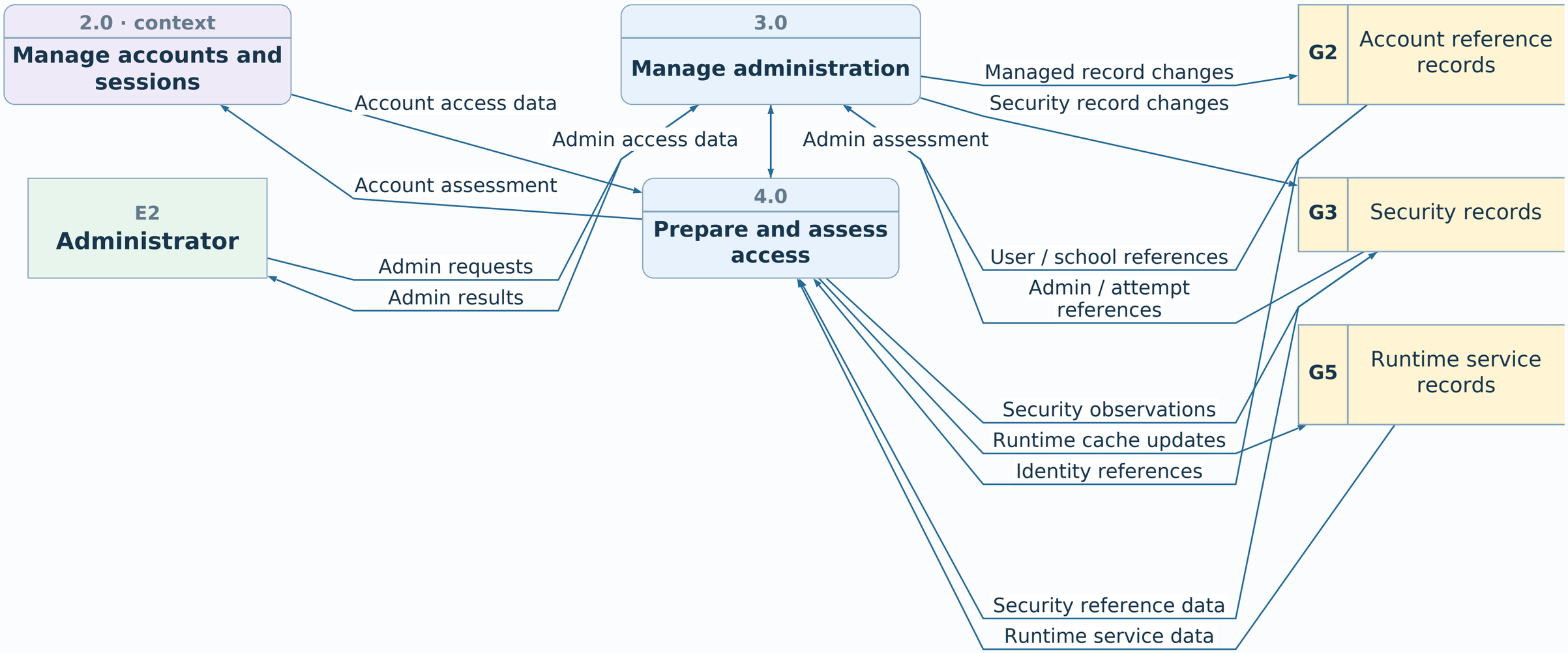


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Grouped stores are expanded into named primitive stores below. Listing API access is independent of the website.

Evidence: Website/src/App.tsx; server/api.mjs, admin.mjs; Backend/Controllers/*, Components/*

Level-0 | Student Exchange data flows | 2/3

Level-0 • Website, Node adapter, backend and logical storage • same-level sheet for 3.0, 4.0

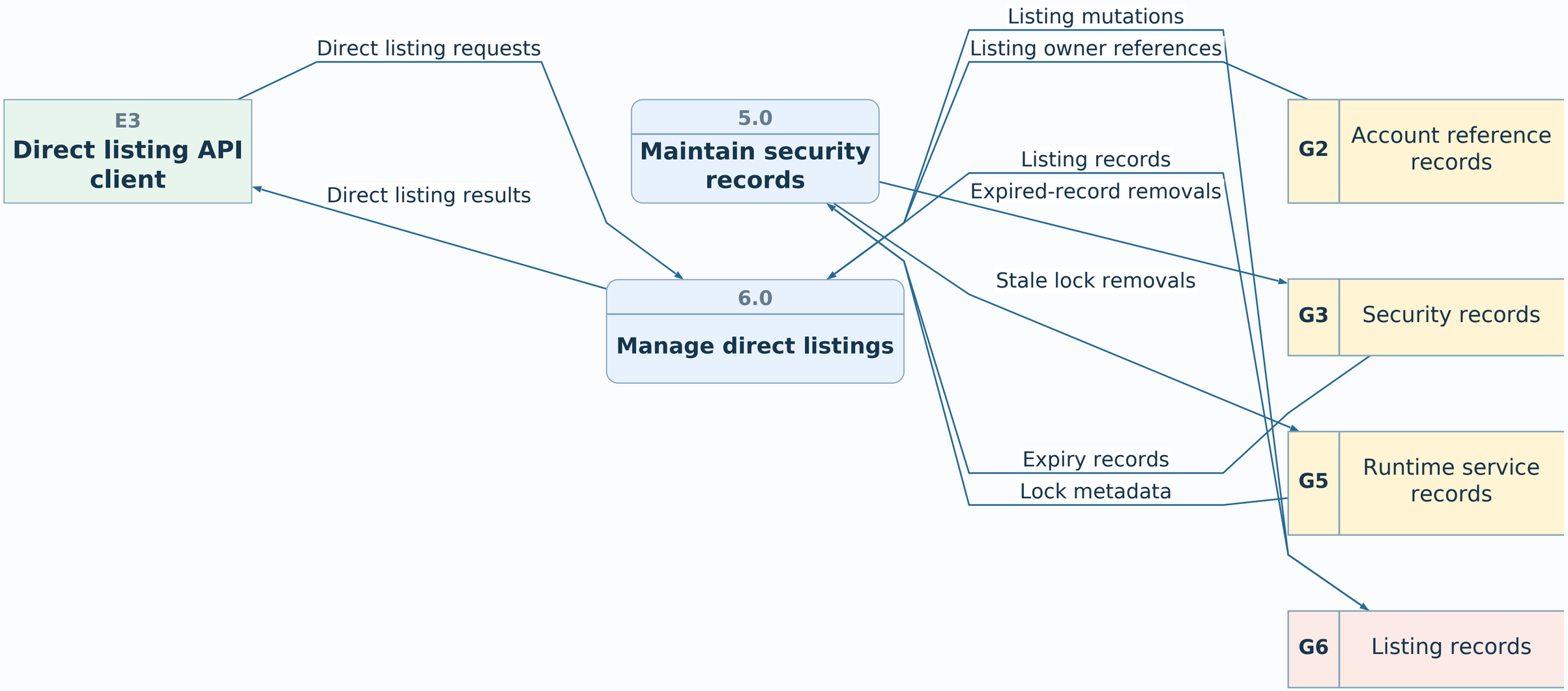


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Grouped stores are expanded into named primitive stores below. Listing API access is independent of the website.

Evidence: Website/src/App.tsx; server/api.mjs, admin.mjs; Backend/Controllers/*, Components/*

Level-0 | Student Exchange data flows | 3/3

Level-0 • Website, Node adapter, backend and logical storage • same-level sheet for 5.0, 6.0

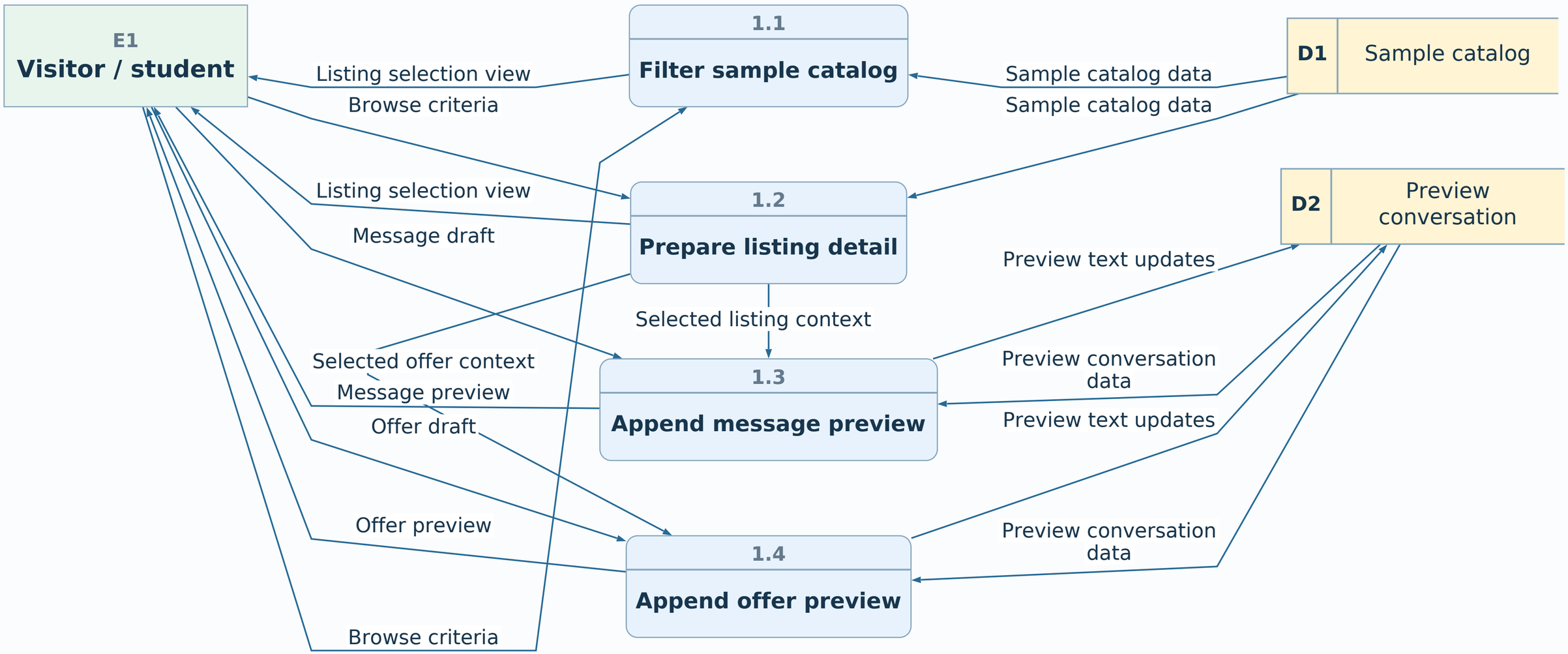


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Grouped stores are expanded into named primitive stores below. Listing API access is independent of the website.

Evidence: Website/src/App.tsx; server/api.mjs, admin.mjs; Backend/Controllers/*, Components/*

1.0 | Marketplace preview data

Level-1 • Decomposition of process 1.0

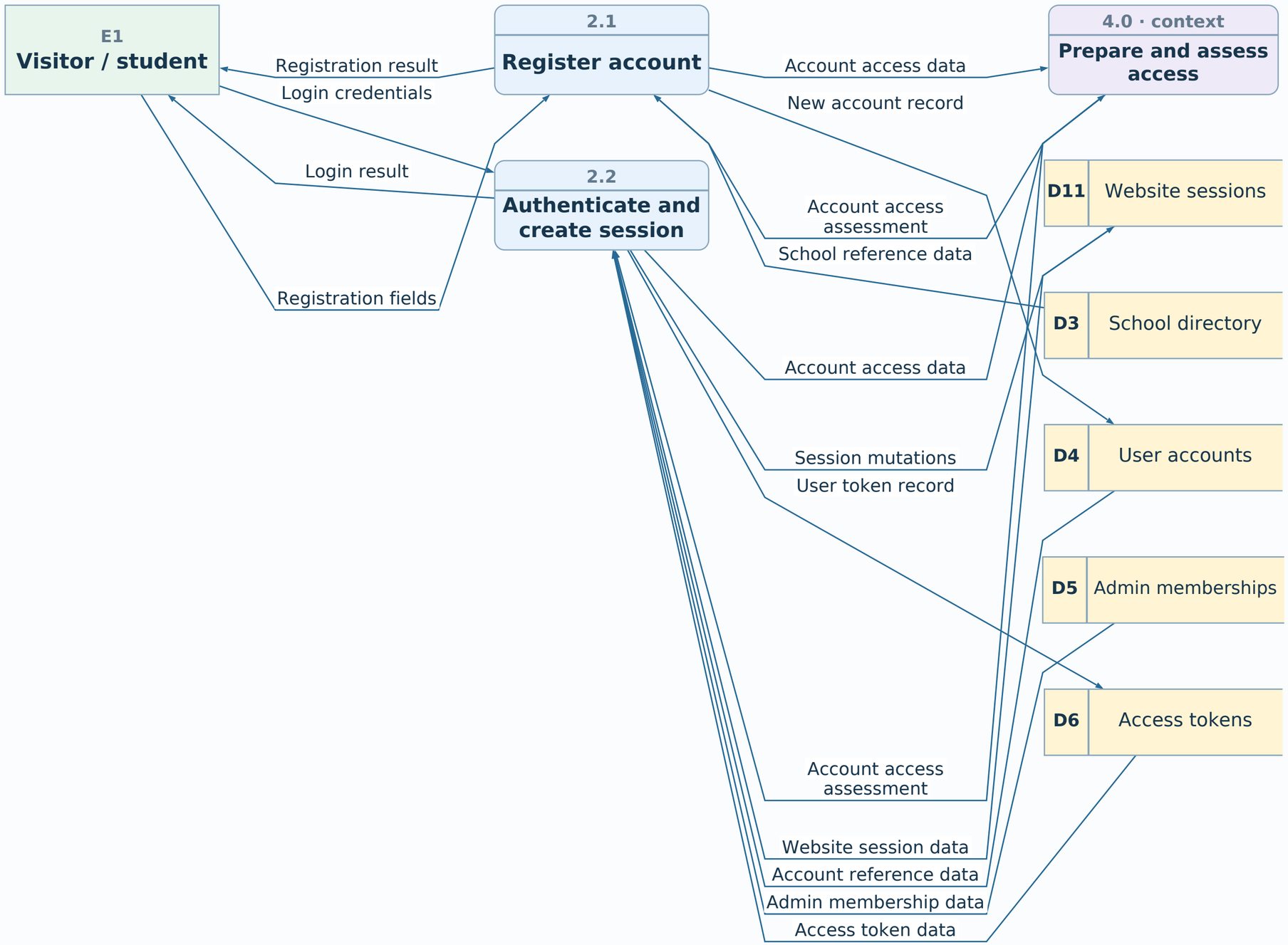


D1 is bundled sample data. D2 is one App-level array, not a per-listing persisted conversation.

Evidence: Website/src/App.tsx: visibleListings and message/offer handlers; src/data/listings.ts

2.0 | Accounts and website sessions | 1/3

Level-1 • Decomposition of process 2.0 • same-level sheet for 2.1, 2.2

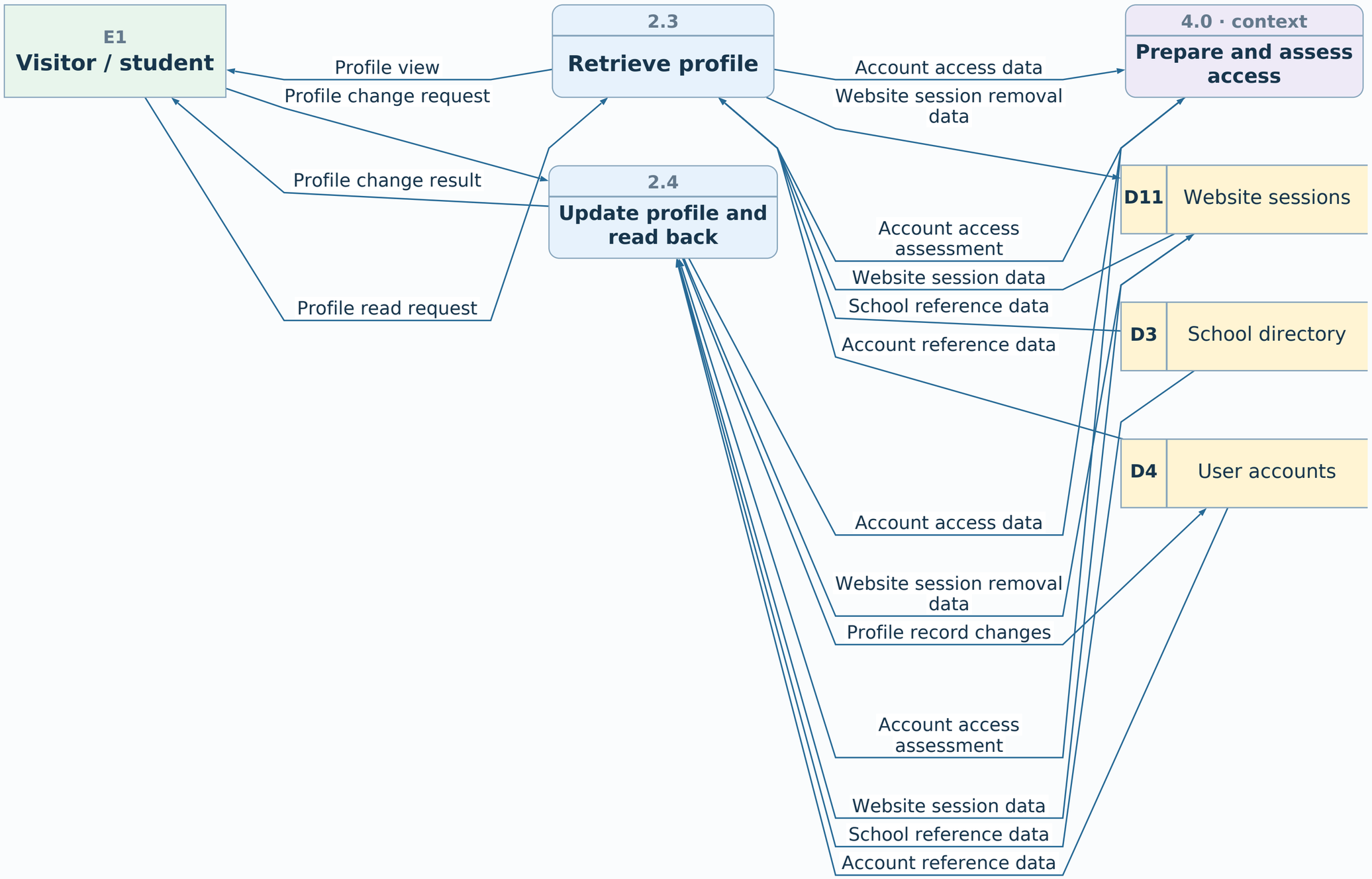


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Shared access flows contain an operation discriminator; parallel-looking arrows represent alternative requests, not execution order.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

2.0 | Accounts and website sessions | 2/3

Level-1 • Decomposition of process 2.0 • same-level sheet for 2.3, 2.4

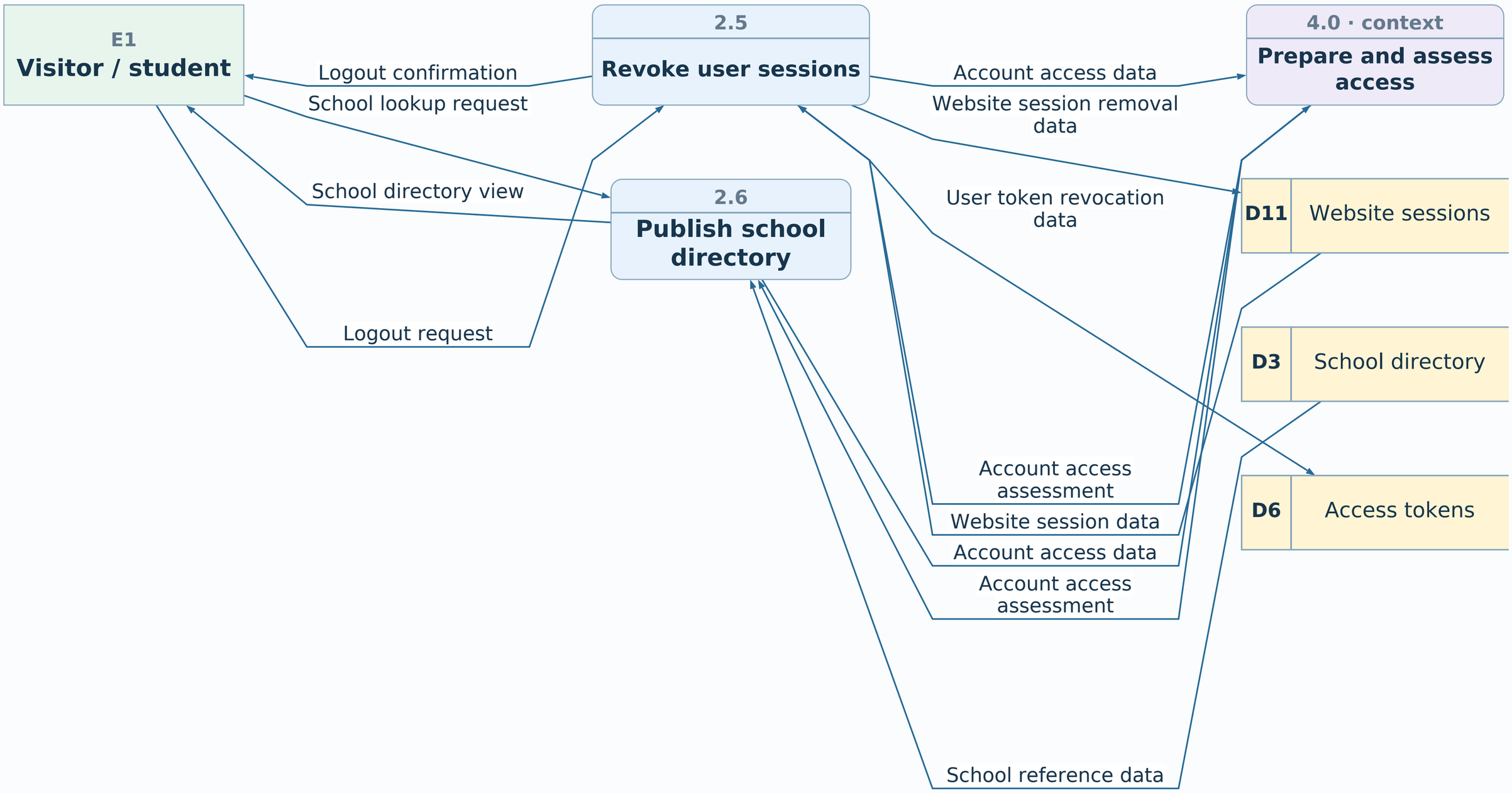


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Shared access flows contain an operation discriminator; parallel-looking arrows represent alternative requests, not execution order.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

2.0 | Accounts and website sessions | 3/3

Level-1 • Decomposition of process 2.0 • same-level sheet for 2.5, 2.6

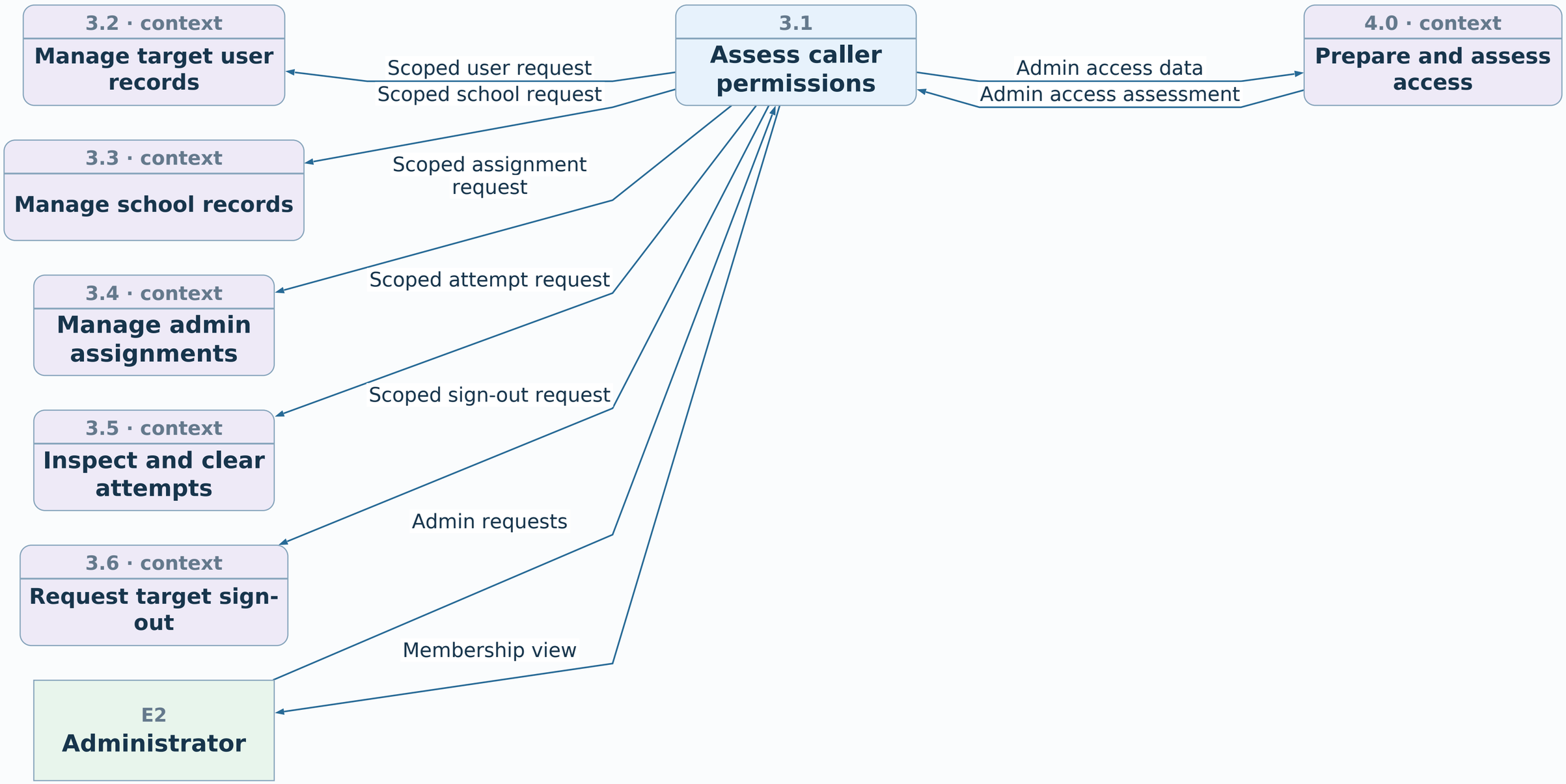


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Shared access flows contain an operation discriminator; parallel-looking arrows represent alternative requests, not execution order.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

3.0 | Administration data | 1/4

Level-1 • Decomposition of process 3.0 • same-level sheet for 3.1

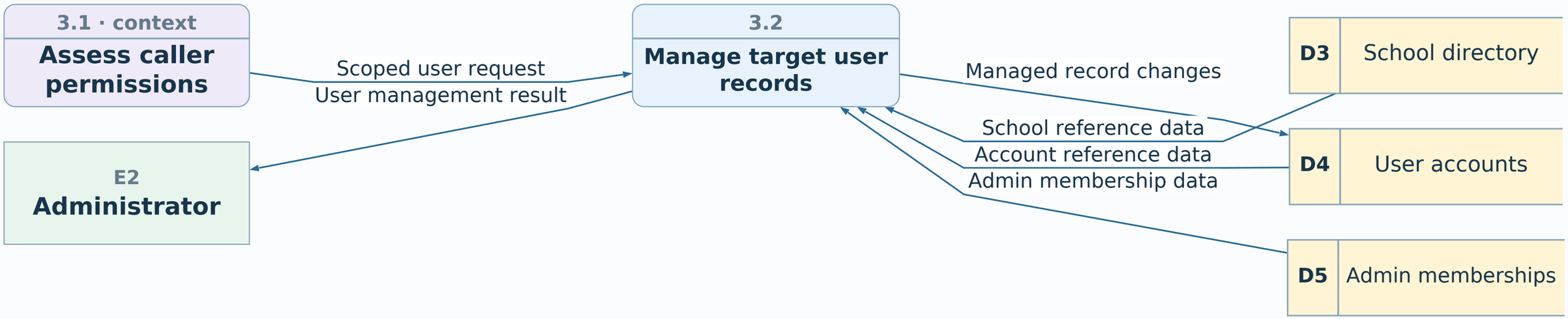


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Fresh membership data qualifies every operation. Target rank/lock checks remain enforced by backend auth.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

3.0 | Administration data | 2/4

Level-1 • Decomposition of process 3.0 • same-level sheet for 3.2

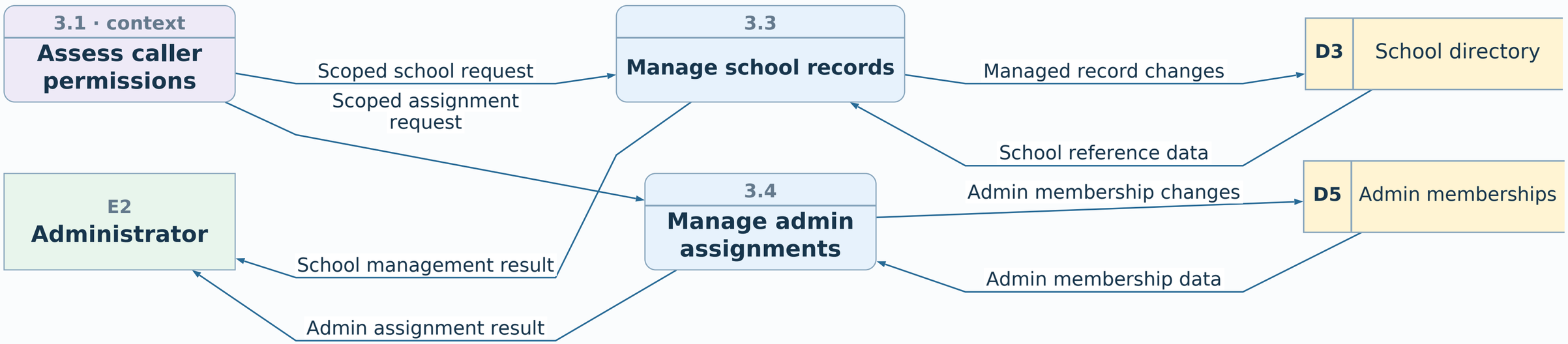


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Fresh membership data qualifies every operation. Target rank/lock checks remain enforced by backend auth.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

3.0 | Administration data | 3/4

Level-1 • Decomposition of process 3.0 • same-level sheet for 3.3, 3.4

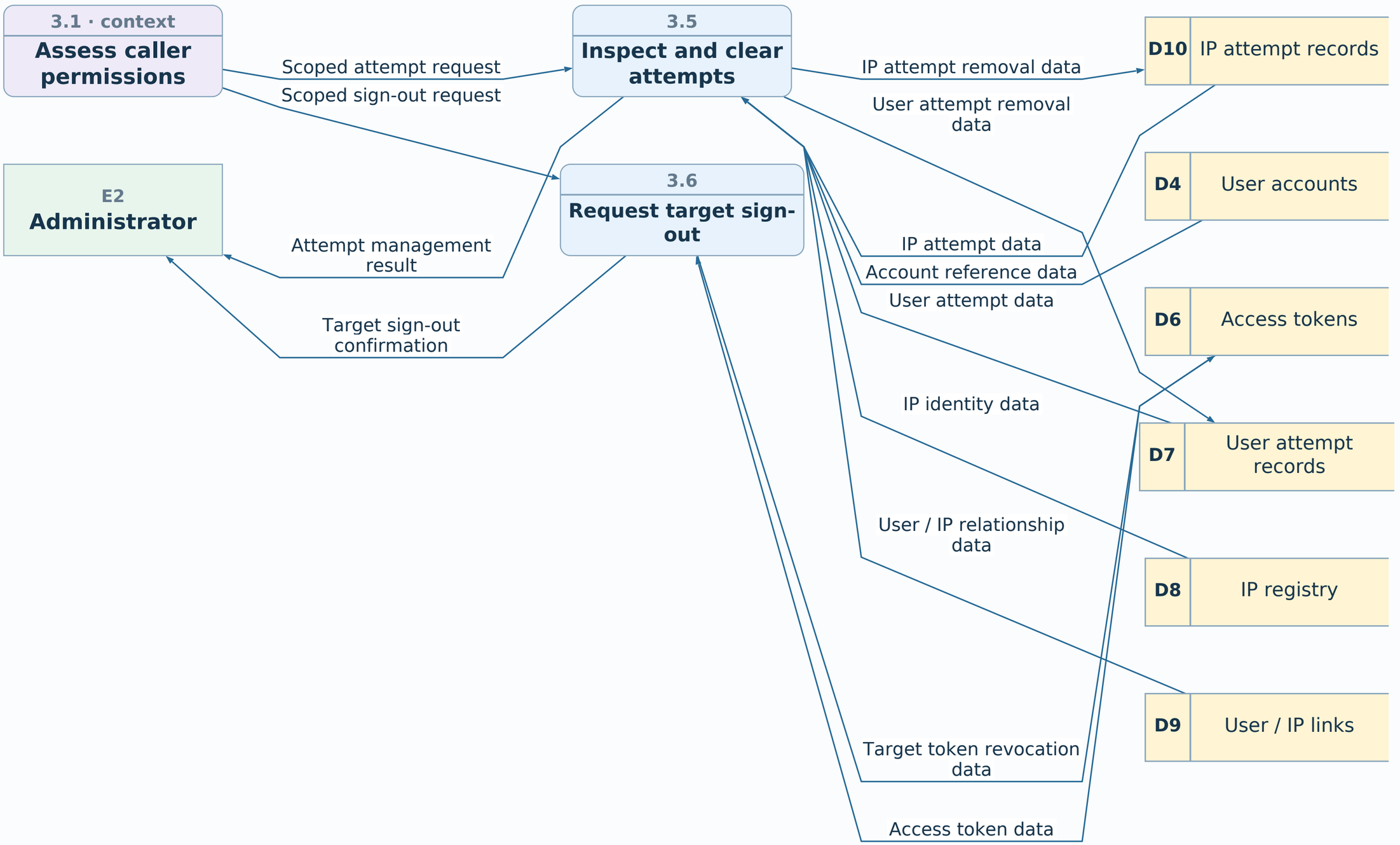


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Fresh membership data qualifies every operation. Target rank/lock checks remain enforced by backend auth.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

3.0 | Administration data | 4/4

Level-1 • Decomposition of process 3.0 • same-level sheet for 3.5, 3.6

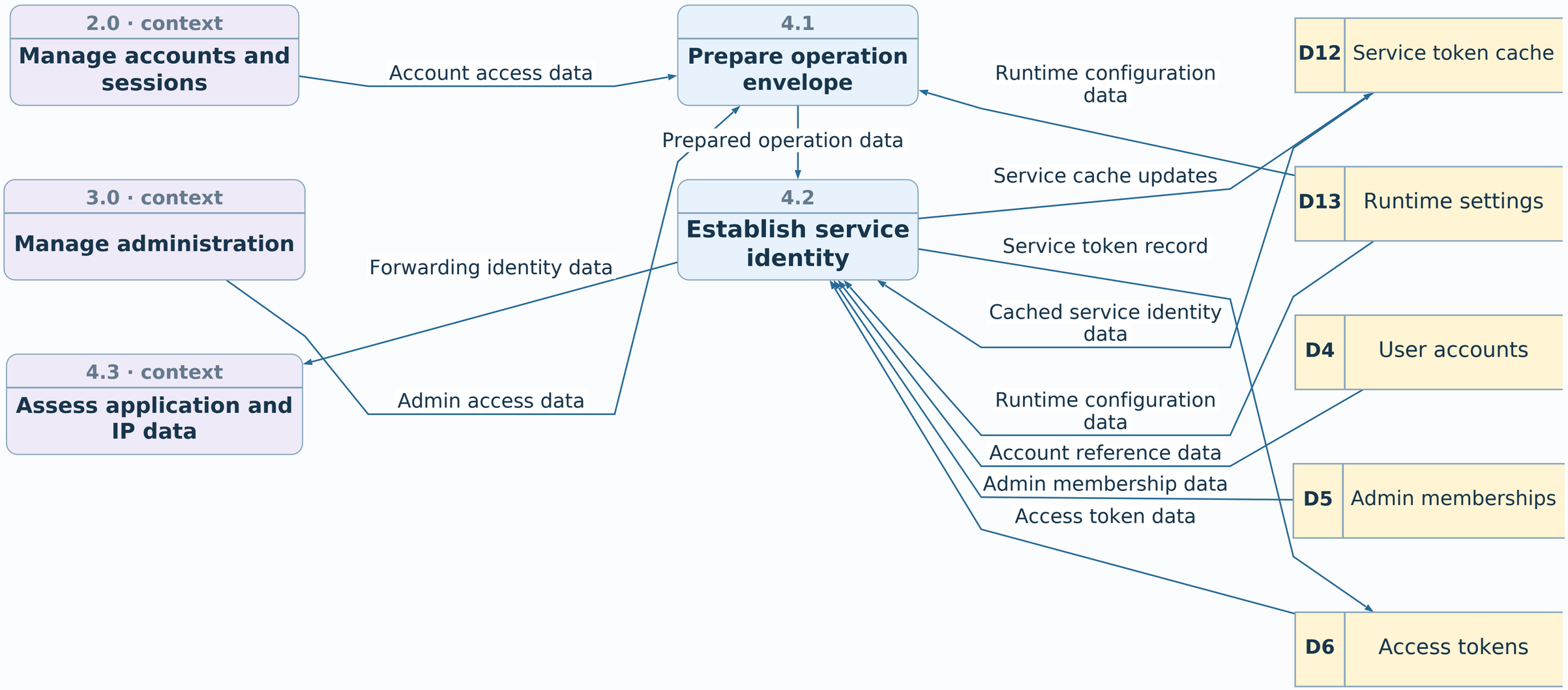


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. Fresh membership data qualifies every operation. Target rank/lock checks remain enforced by backend auth.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

4.0 | Access and forwarding data | 1/4

Level-1 • Decomposition of process 4.0 • same-level sheet for 4.1, 4.2

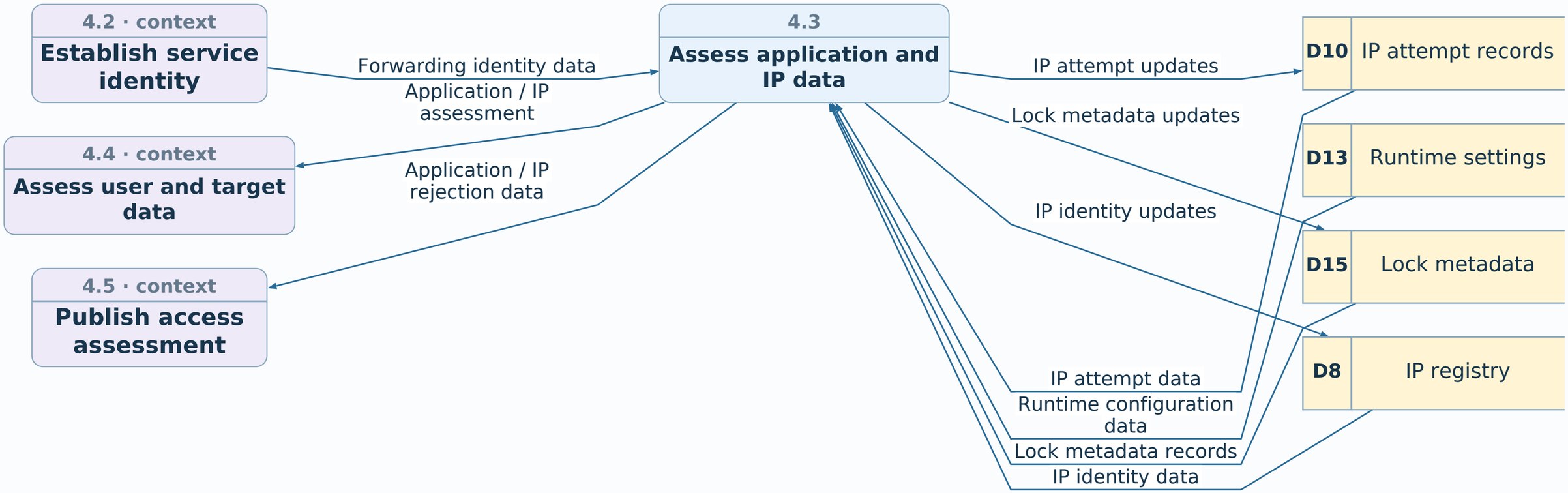


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. This logical transformation spans the Node adapter and backend helpers; it is not a separate deployed service.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

4.0 | Access and forwarding data | 2/4

Level-1 • Decomposition of process 4.0 • same-level sheet for 4.3

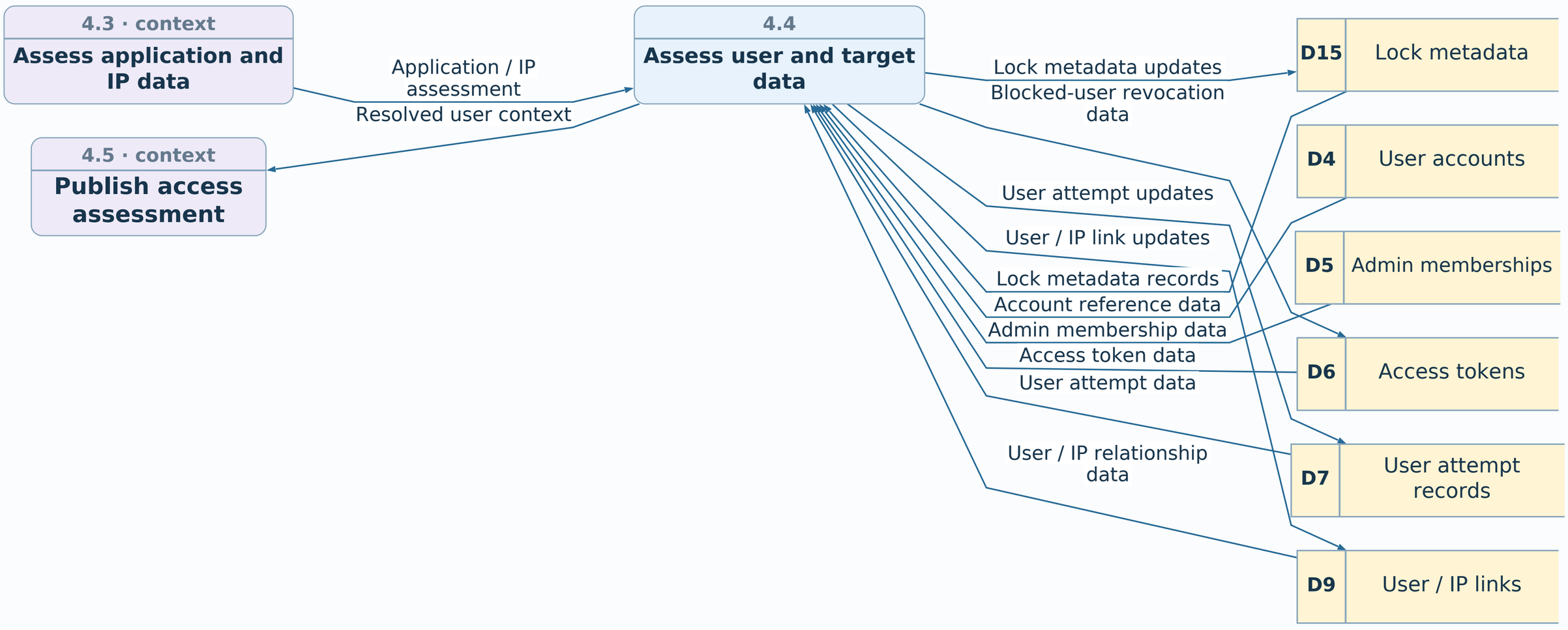


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. This logical transformation spans the Node adapter and backend helpers; it is not a separate deployed service.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

4.0 | Access and forwarding data | 3/4

Level-1 • Decomposition of process 4.0 • same-level sheet for 4.4

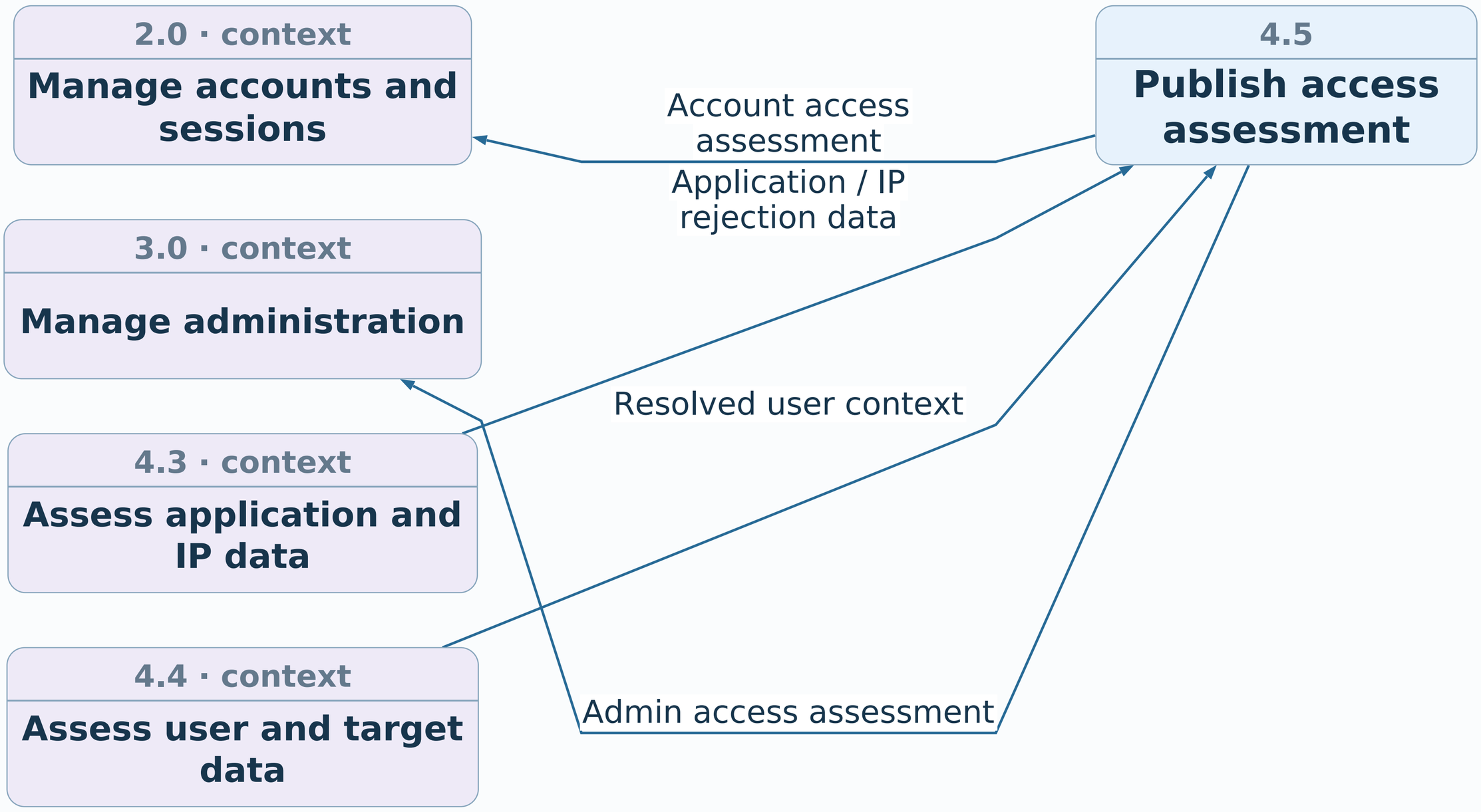


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. This logical transformation spans the Node adapter and backend helpers; it is not a separate deployed service.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

4.0 | Access and forwarding data | 4/4

Level-1 • Decomposition of process 4.0 • same-level sheet for 4.5

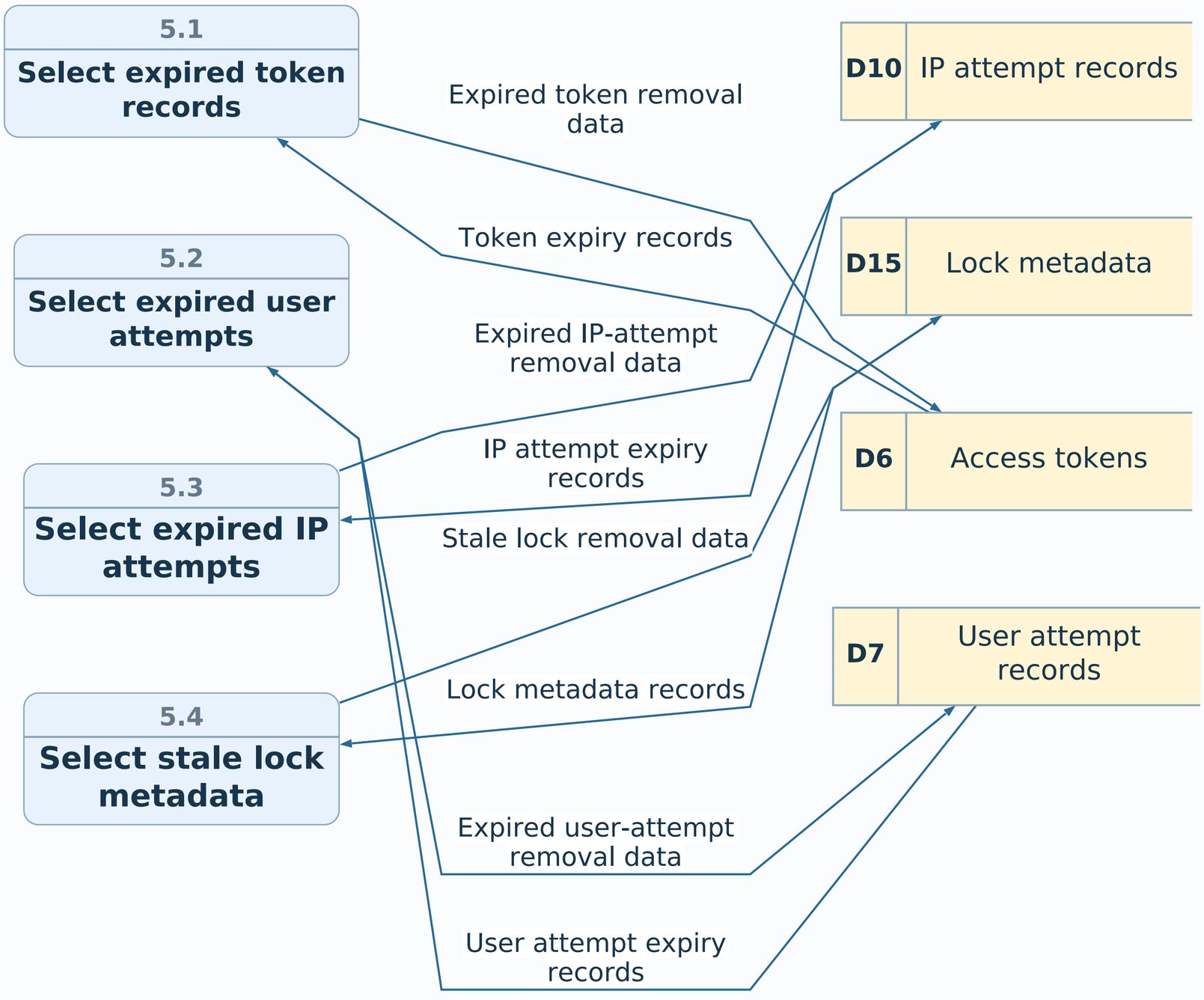


Same-level sheet: purple context processes refer to unchanged IDs elsewhere. This logical transformation spans the Node adapter and backend helpers; it is not a separate deployed service.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

5.0 | Retained security data maintenance

Level-1 • Decomposition of process 5.0

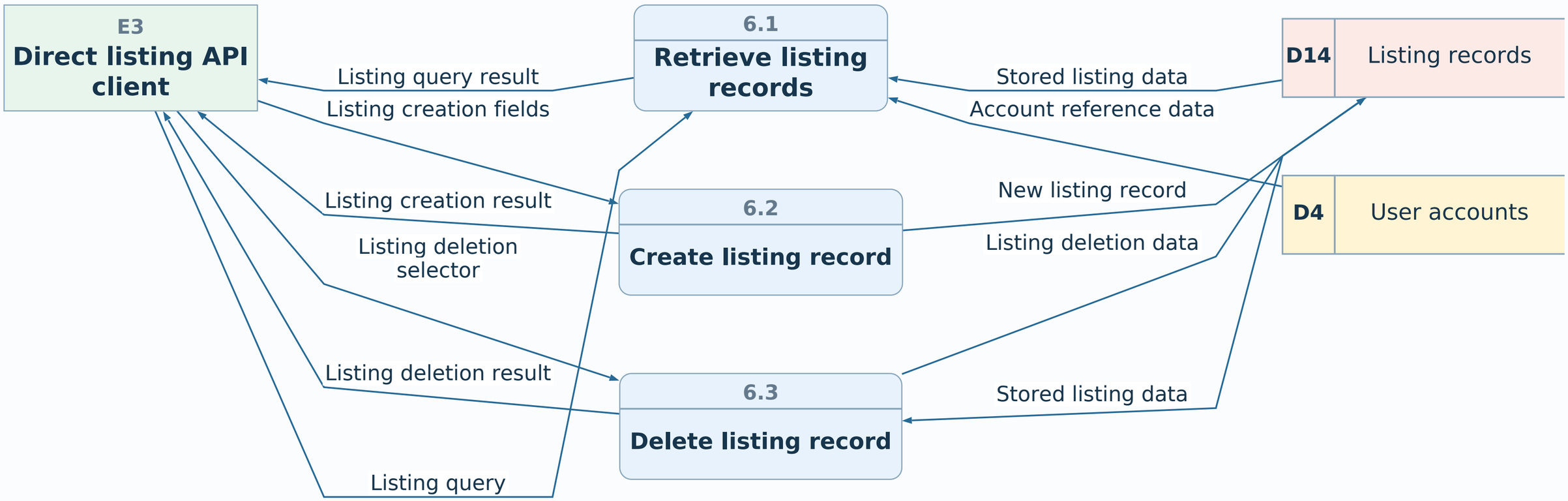


Timing is a process specification, not a data-flow arrow: asynchronous startup cleanup and daily 03:24 Mountain work.

Evidence: Backend/Program.cs; LogIn.CleanUp; Users.CleanUp; IP.CleanUp; Maitenance.cs

6.0 | Direct listing API data

Level-1 • Decomposition of process 6.0

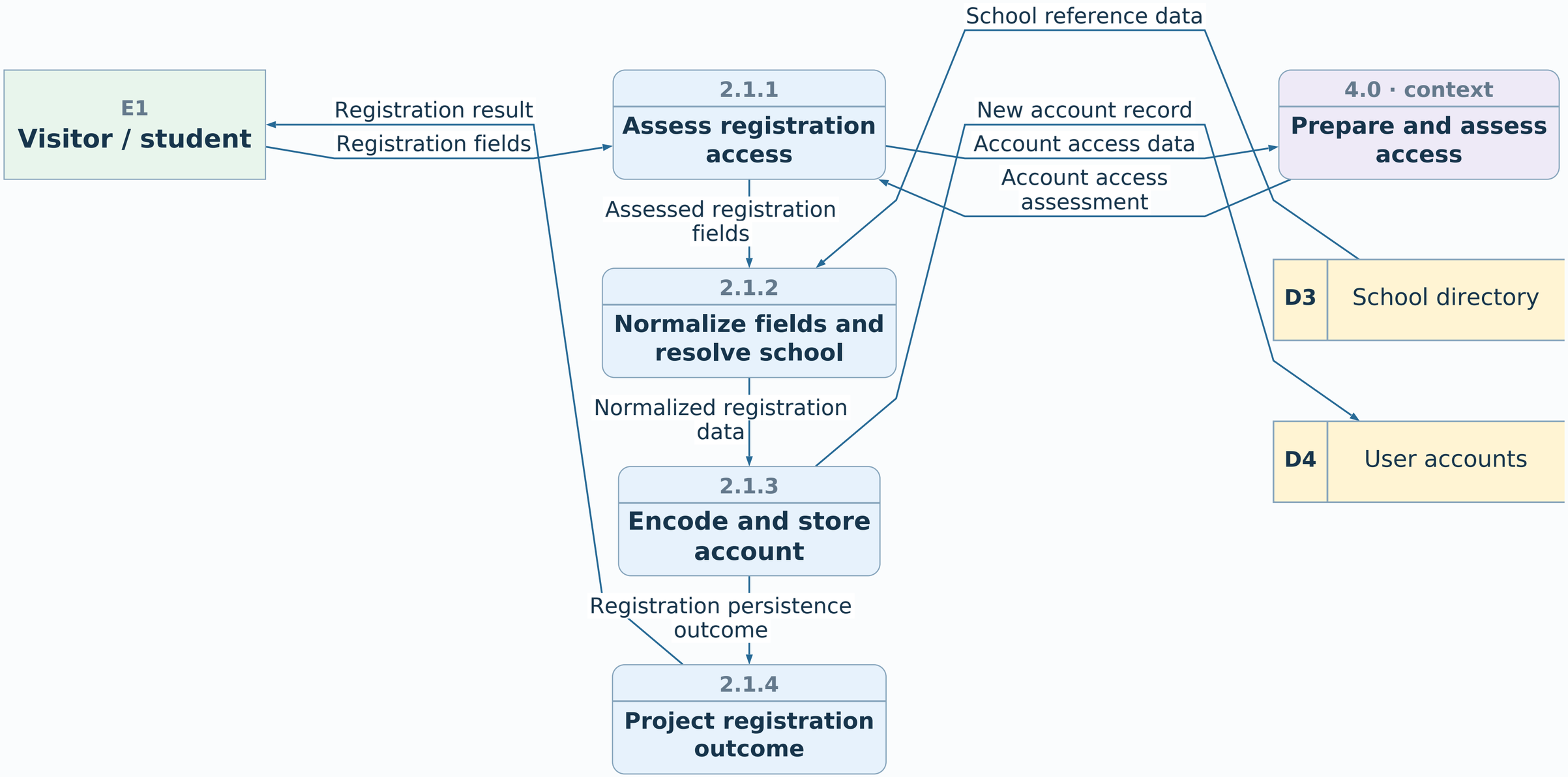


Direct API client only. No website listing adapter route, shared auth/ownership checks or supplied Listings DDL.

Evidence: Backend/Controllers/Listing.cs; DB_Schema.txt; Website/server/api.mjs route inventory

2.1 | Registration data transformation

Level-2 • Decomposition of process 2.1

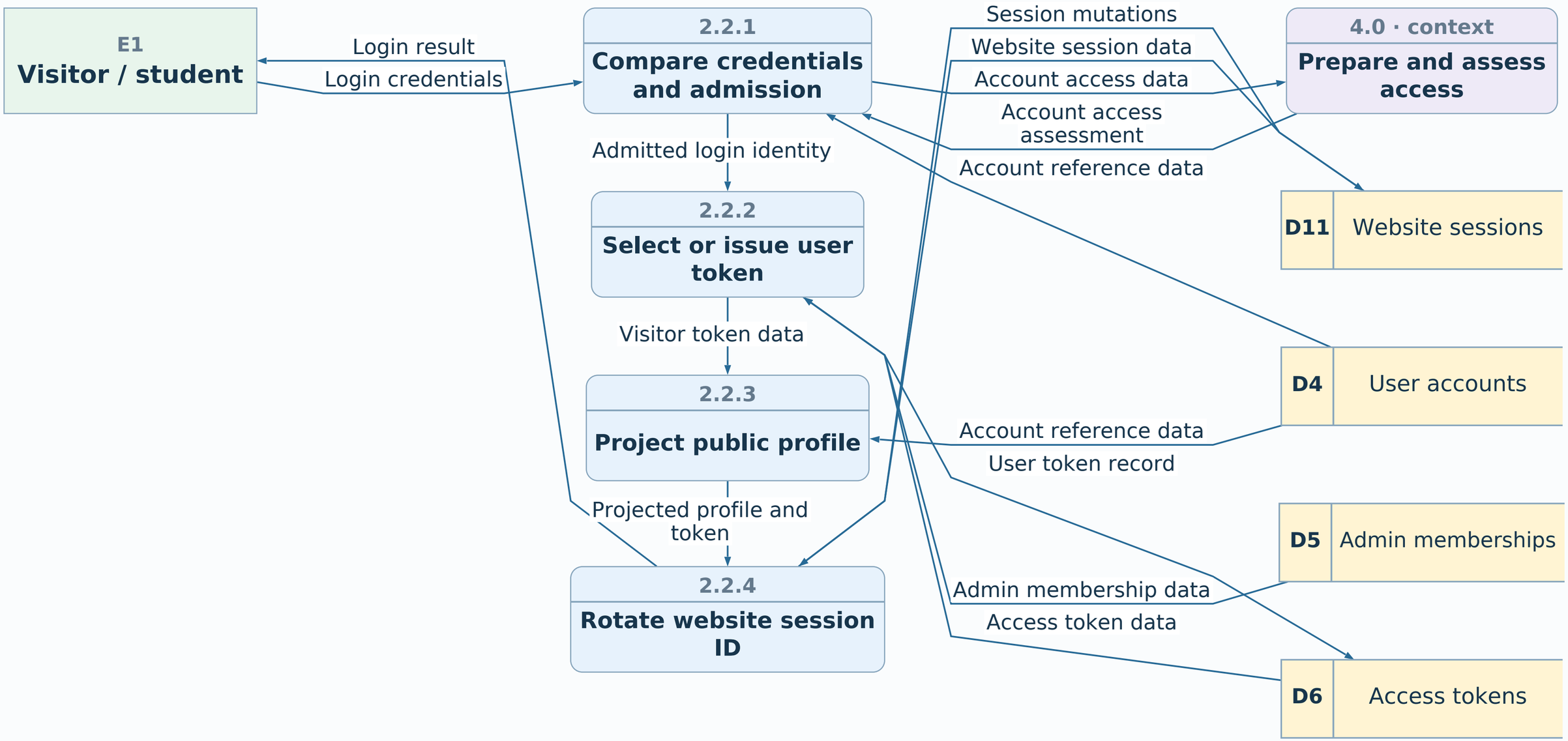


Registration does not issue an access token or create a website session.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

2.2 | Login token and cookie data

Level-2 • Decomposition of process 2.2

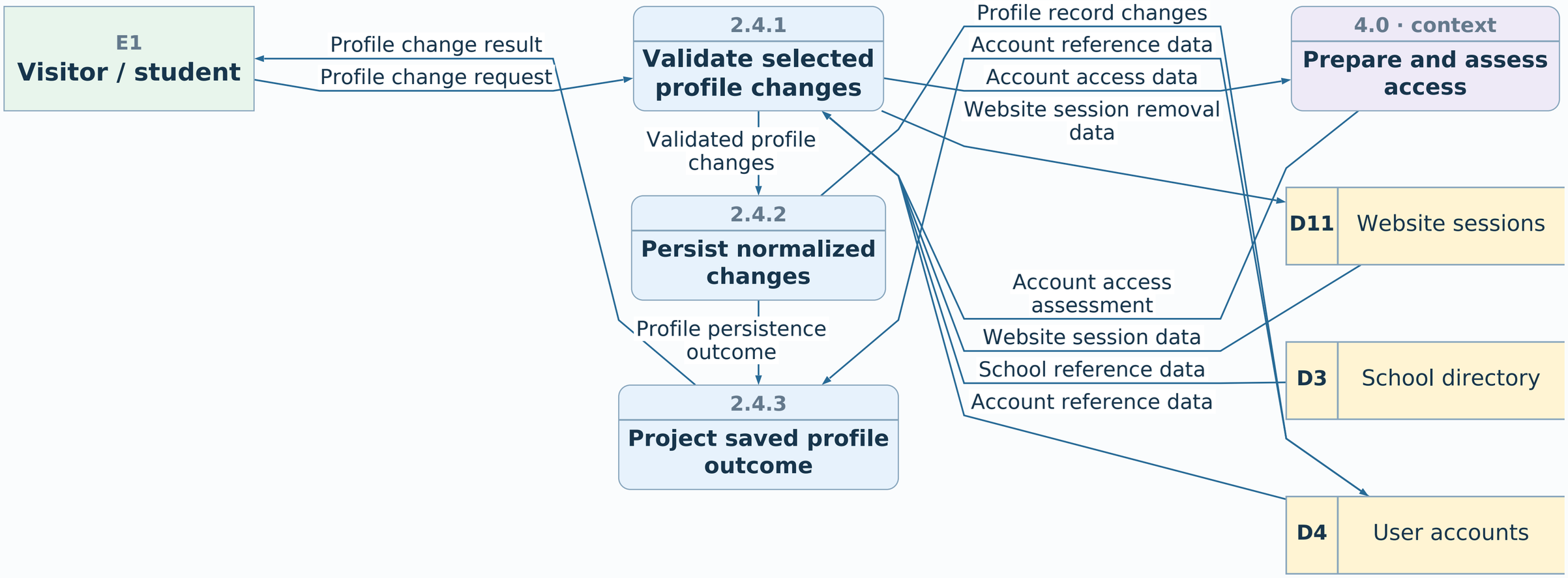


Backend token lifetime: 21 days, or 3 for admins. Website session: fixed 12 hours. Browser sees only an opaque ID.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

2.4 | Saved profile and readback data

Level-2 • Decomposition of process 2.4

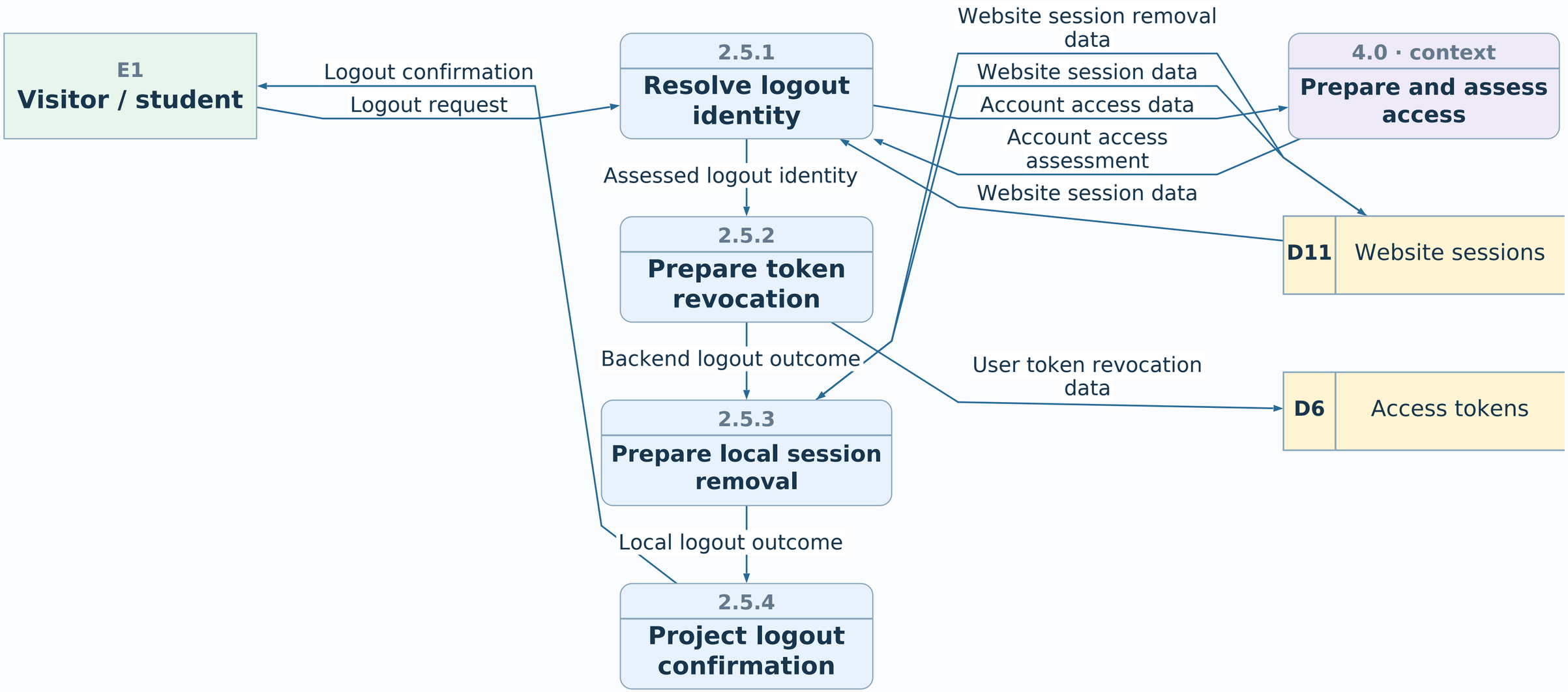


A committed write plus failed readback yields a saved notice and null profile; it does not imply a failed write.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

2.5 | Logout and revocation data

Level-2 • Decomposition of process 2.5

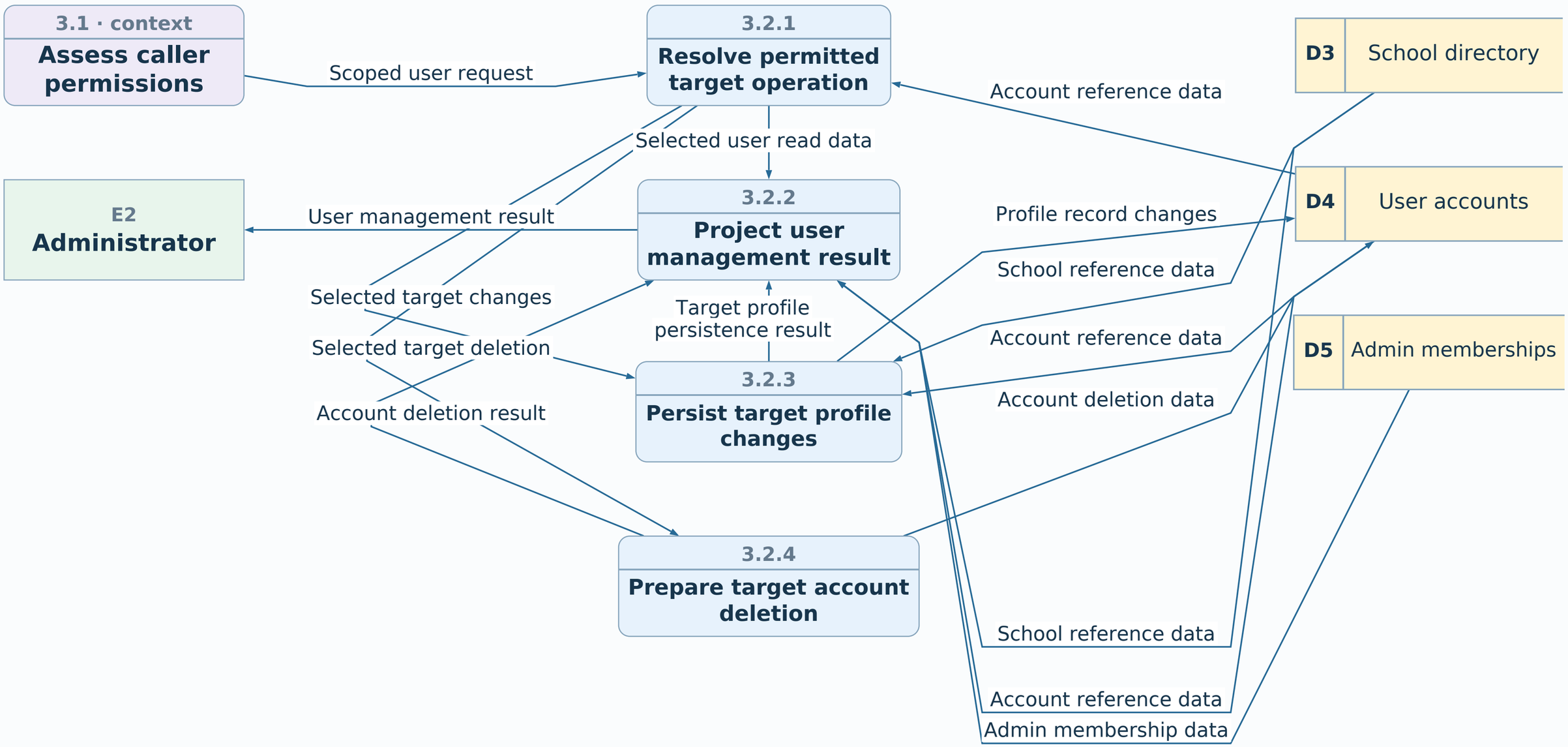


Backend 200 can mask token-validation failure. A confirmation is not evidence that target tokens existed or were deleted.

Evidence: Website/server/api.mjs: account routes; Backend/Register.cs, LogIn.cs, Schools.cs

3.2 | Target user record management

Level-2 • Decomposition of process 3.2

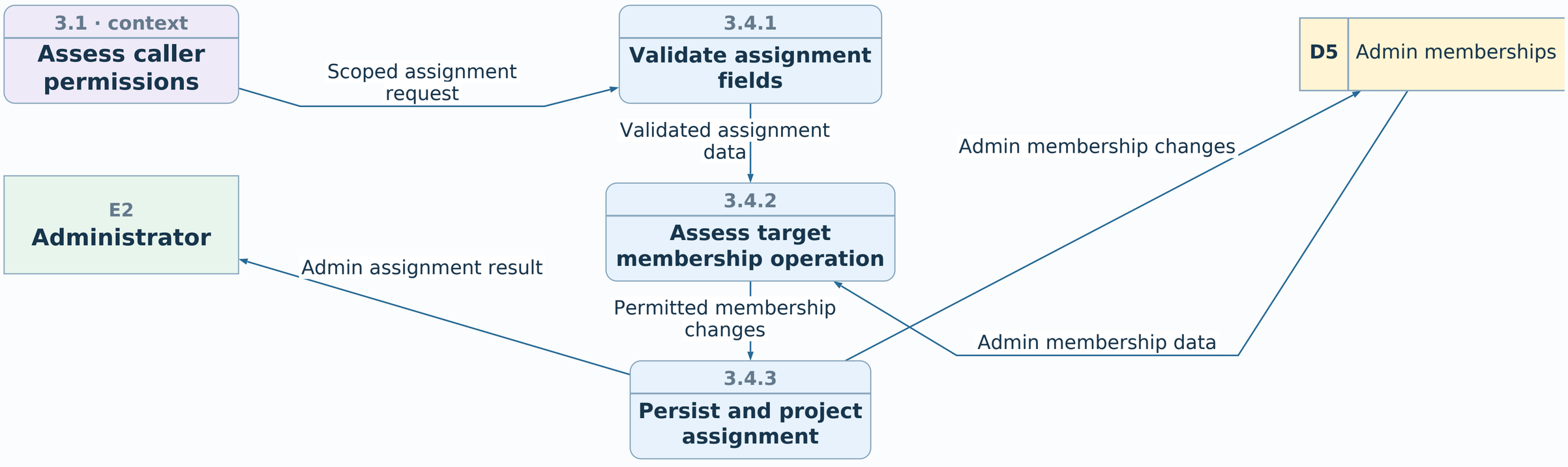


Caller access_token remains the admin token. to_user selects the target and must match the stored normalized email exactly.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

3.4 | Admin membership changes

Level-2 • Decomposition of process 3.4

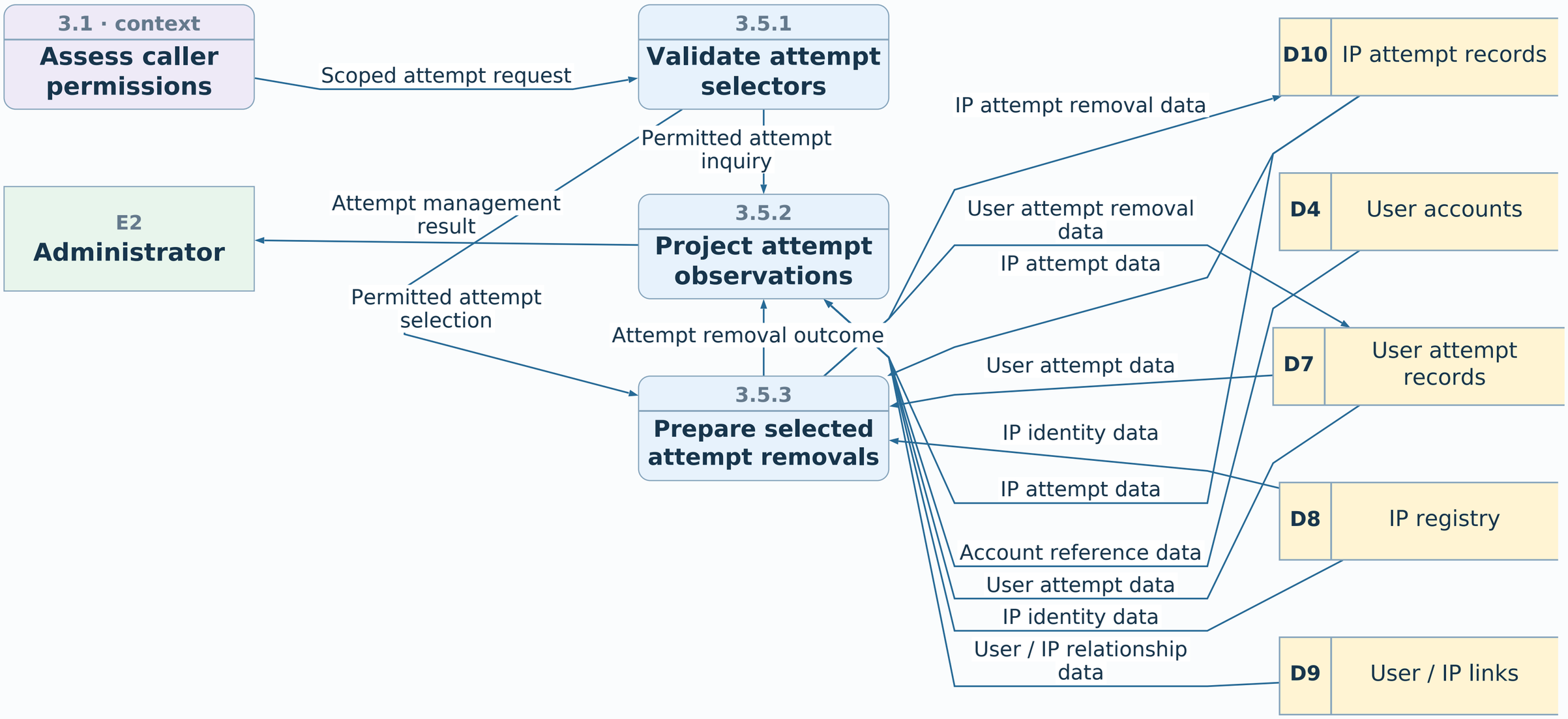


Admin management needs caller level > 0; explicitly setting locked needs level ≥ 30. Target admins must have lower rank.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

3.5 | Attempt inspection and clearing

Level-2 • Decomposition of process 3.5

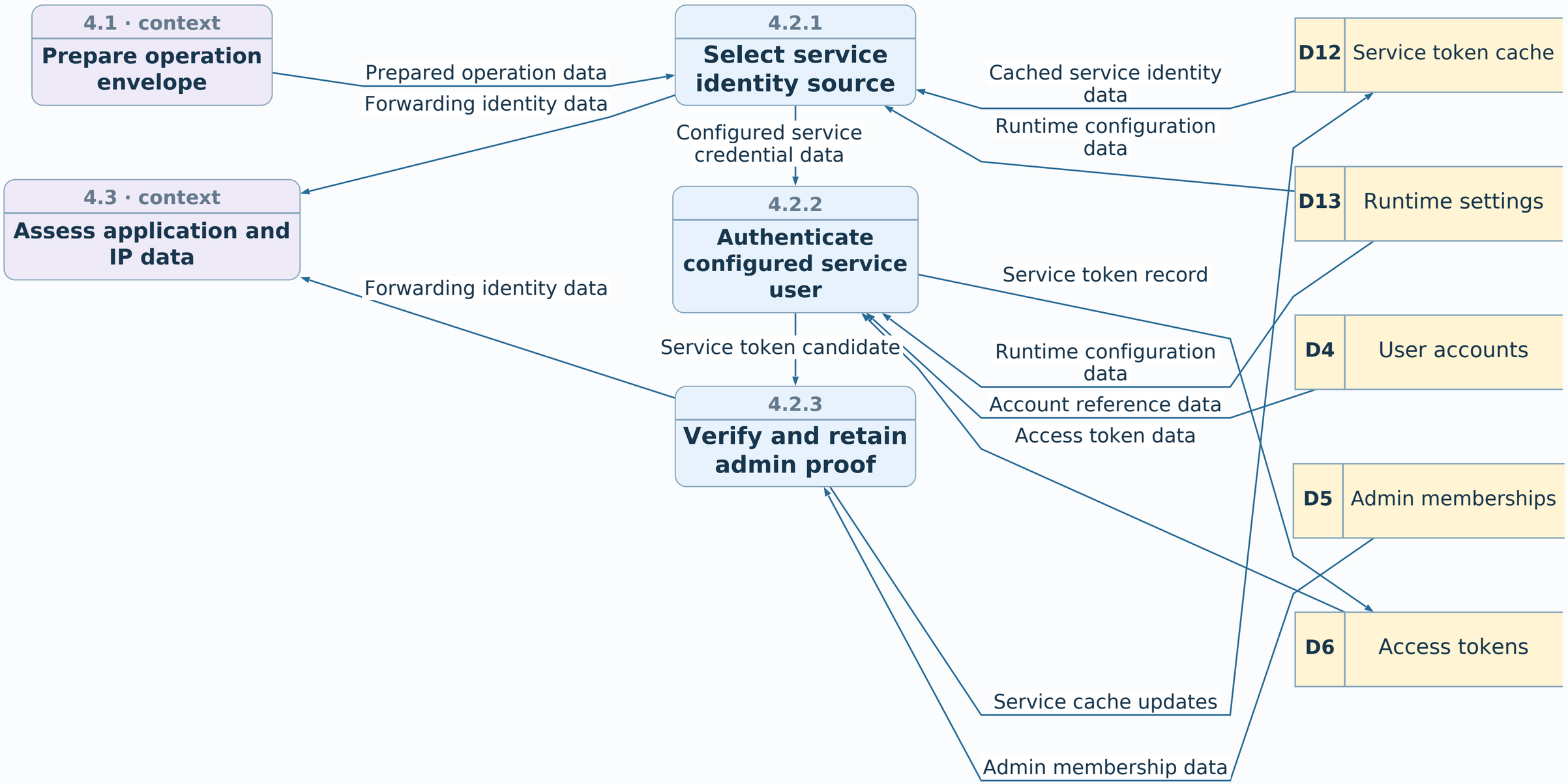


Rows include observations below the block threshold; clearing records does not recreate revoked access tokens.

Evidence: Website/server/admin.mjs, api.mjs; Backend/Admin.cs, Register.cs, Schools.cs, Blocked.cs, Users.cs

4.2 | Forwarding identity data

Level-2 • Decomposition of process 4.2

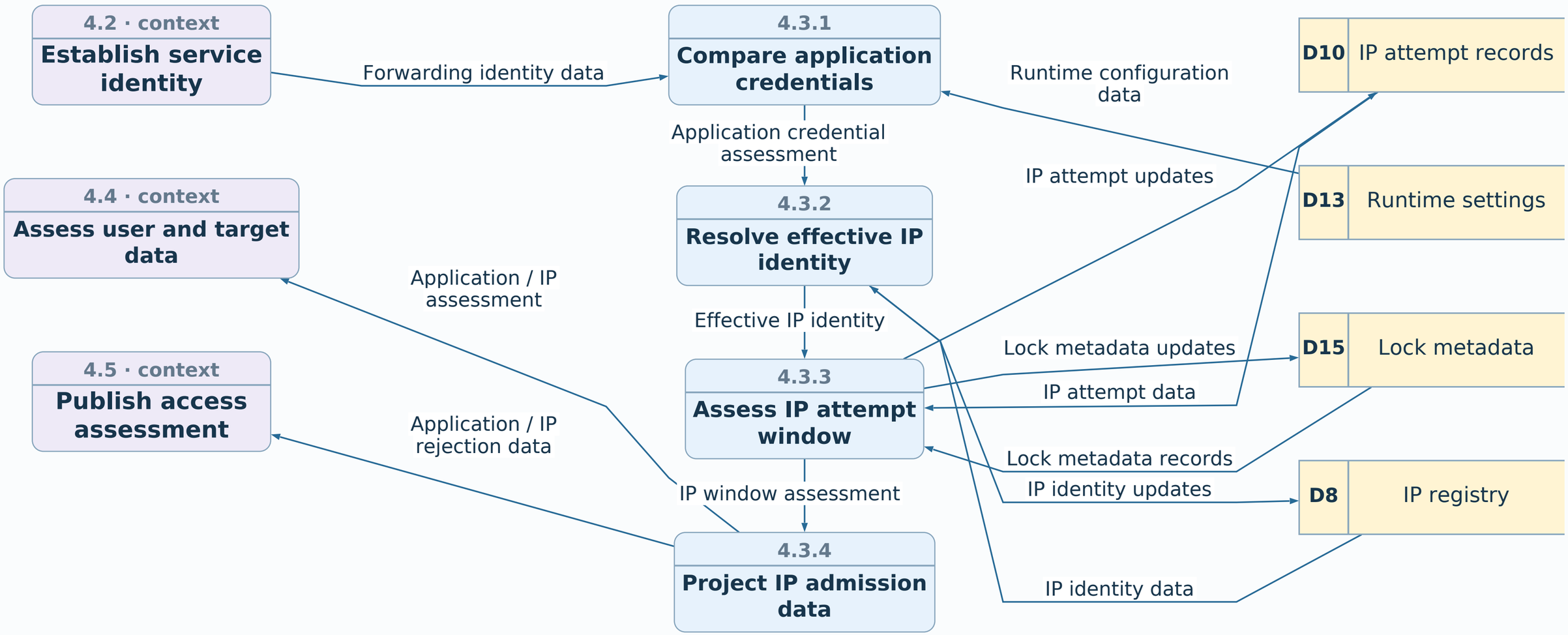


Service admin_token and visitor access_token are different data. Only explicit forwarding-expiry rejection permits one renewal retry.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

4.3 | Application and IP observations

Level-2 • Decomposition of process 4.3

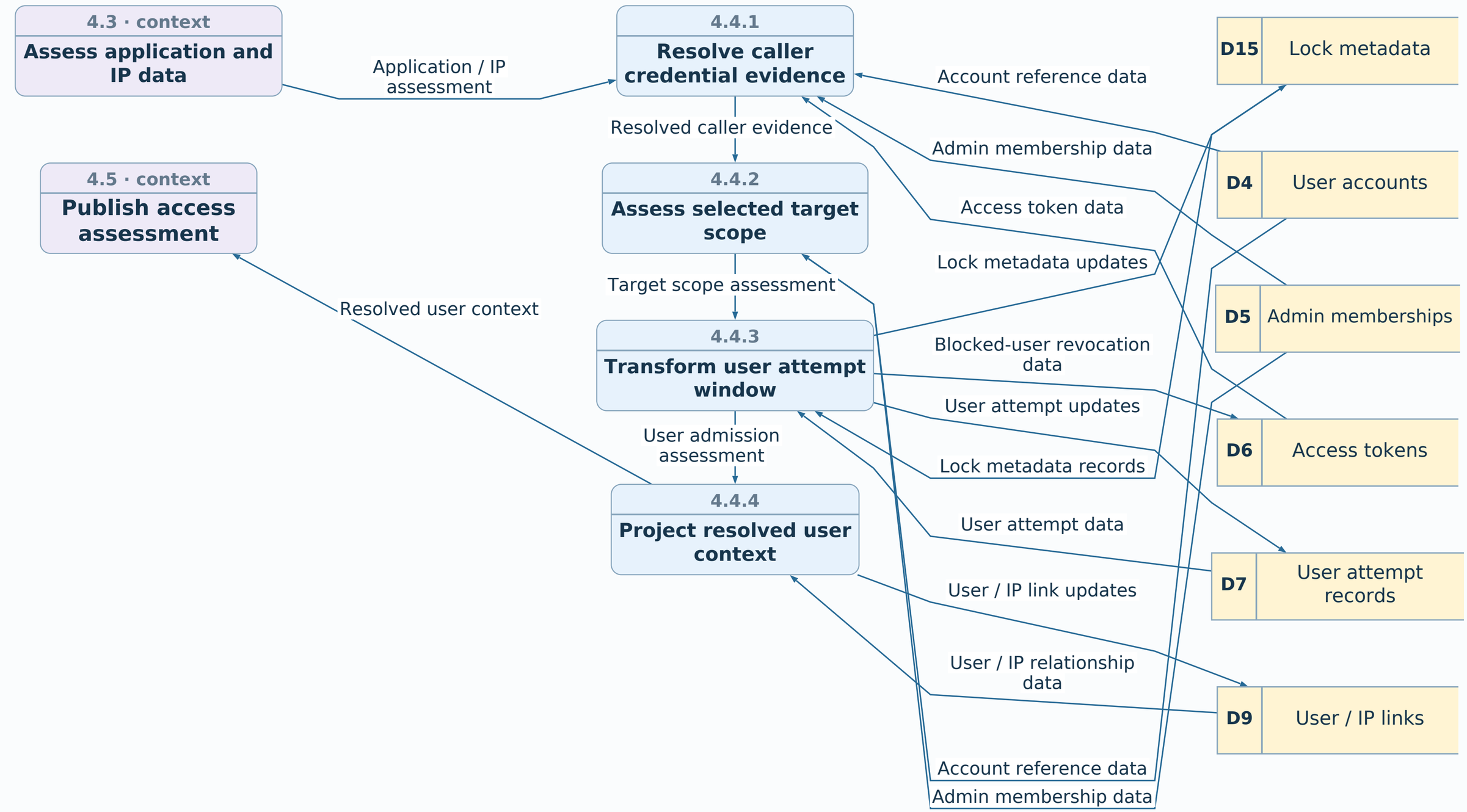


IP rows contain encrypted addresses. Forwarded-IP trust is server-owned; browser fields cannot replace the security envelope.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

4.4 | User identity and attempt observations

Level-2 • Decomposition of process 4.4



Attempt blocking is applied to login admission; ordinary token validation does not itself consult Blocked_Users.

Evidence: Website/server/api.mjs: forwarding client, upstream; Backend/AppAuthJson.cs, Users.cs, IP.cs

Boundary conservation audit

Automated comparison of directional data-package sets, including grouped-store expansions.

Child view	Parent process	Input packages	Output packages	Result
L0	0	18	18	**PASS**
P1	1.0	5	4	**PASS**
P2	2.0	12	13	**PASS**
P3	3.0	15	15	**PASS**
P4	4.0	12	10	**PASS**
P5	5.0	4	4	**PASS**
P6	6.0	5	5	**PASS**
P21	2.1	3	3	**PASS**
P22	2.2	6	5	**PASS**
P24	2.4	5	4	**PASS**
P25	2.5	3	4	**PASS**
P32	3.2	4	3	**PASS**
P34	3.4	2	2	**PASS**
P35	3.5	6	3	**PASS**
P42	4.2	6	3	**PASS**
P43	4.3	5	5	**PASS**
P44	4.4	7	5	**PASS**

Balancing preserves external information, not the number of arrows. Internal flows may be added within a child.

Reference: DFD (1).pdf, pp. 19–23; validation-report.json contains the machine-readable results.

Logical and physical data store dictionary

Grouped Level-0 stores are refined into primitive subject stores on child diagrams.

D1	**Sample catalog**	**G1**	src/data/listings.ts · bundled constant
D2	**Preview conversation**	**G1**	App.tsx · messages[] in browser memory
D3	**School directory**	**G2**	SQLite Schools
D4	**User accounts**	**G2**	SQLite User
D5	**Admin memberships**	**G3**	SQLite Admins
D6	**Access tokens**	**G3**	SQLite Access_Token
D7	**User attempt records**	**G3**	SQLite Blocked_Users
D8	**IP registry**	**G3**	SQLite IP
D9	**User / IP links**	**G3**	SQLite User_Has_IPs
D10	**IP attempt records**	**G3**	SQLite Blocked_IPs
D11	**Website sessions**	**G4**	Node sessions Map · opaque ID → token / expiry
D12	**Service token cache**	**G5**	Node forwarding-client closure
D13	**Runtime settings**	**G5**	Server environment + backend Env policy / paths
D14	**Listing records**	**G6**	Listings table inferred from SQL; missing from DDL
D15	**Lock metadata**	**G5**	Users._userLock and IP lock dictionaries

Open-ended stores represent retained information, including process memory; they do not all imply database tables.

Reference: DFD (1).pdf, pp. 3–4, 15–23 and 27; full fields and retention notes in DATA-DICTIONARY.md.

Current data boundaries and implementation gaps

The diagrams describe existing behavior and clearly qualify incomplete listing storage.

Preview data stays in the browser

Messages and offer strings update one App-level array. They are not sent to sellers and are not isolated by listing.

Two identities cross the adapter boundary

Service admin_token enables server-owned IP forwarding; visitor access_token identifies the acting user. The browser receives an opaque cookie ID.

Listing storage is a qualified expectation

ListingController uses ../Data.db and expects Listings, which is absent from supplied DDL. The website has no listing adapter route.

Retention is not a scheduling arrow

Node sessions disappear on restart. Security maintenance uses expiry data. Startup cleanup is unawaited; daily cleanup is at 03:24 Mountain.

Logical processes span implementation modules

Process 4.0 centralizes access-related data transformations across Node and ASP.NET. It does not assert a separate physical service.

No payment, completed order, seller delivery or persisted offer acceptance is invented in these DFDs.

Evidence: Website/App.tsx, data/listings.ts, server/api.mjs; Backend/Listing.cs, Sql.cs, Env.cs.